

- 1** A particle P of mass m is attached to one end of a light inextensible string of length a . The other end of the string is attached to a fixed point O . The string is held taut with OP making an angle α with the downward vertical, where $\cos \alpha = \frac{2}{3}$. The particle P is projected perpendicular to OP in an upwards direction with speed $\sqrt{3ag}$. It then starts to move along a circular path in a vertical plane.

Find the cosine of the angle between the string and the upward vertical when the string first becomes slack. [4]

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.